

WASP-104b

R-2598

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-104_160301 · created 2026-08-28T00:08:00+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457448.822806 +/- 7e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.029 +/- 0.038
Transit depth (Rp/Rs)^2	0.086 +/- 0.22 [%]
Orbital Inclination (inc)	85.06 +/- 2.88
Airmass coefficient 1 (a1)	1.0101 +/- 0.0078
Airmass coefficient 2 (a2)	-0.008 +/- 0.0063
Scatter in the residuals of the lightcurve fit is	0.77 %
Best Comparison Star	#2 - [47, 234]
Optimal Aperture	3.43
Optimal Annulus	4.08
Transit Duration (day)	0.067 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.029 $\pm$ 0.038 vs 0.1214 (deviation 2.43 σ)
Duration fit vs published	0.067 d vs 0.0742 d (deviation 0.29 σ)
Mid-time O-C	-0.1 min
Depth SNR	0.8
Residual scatter	0.77 %

Light curve

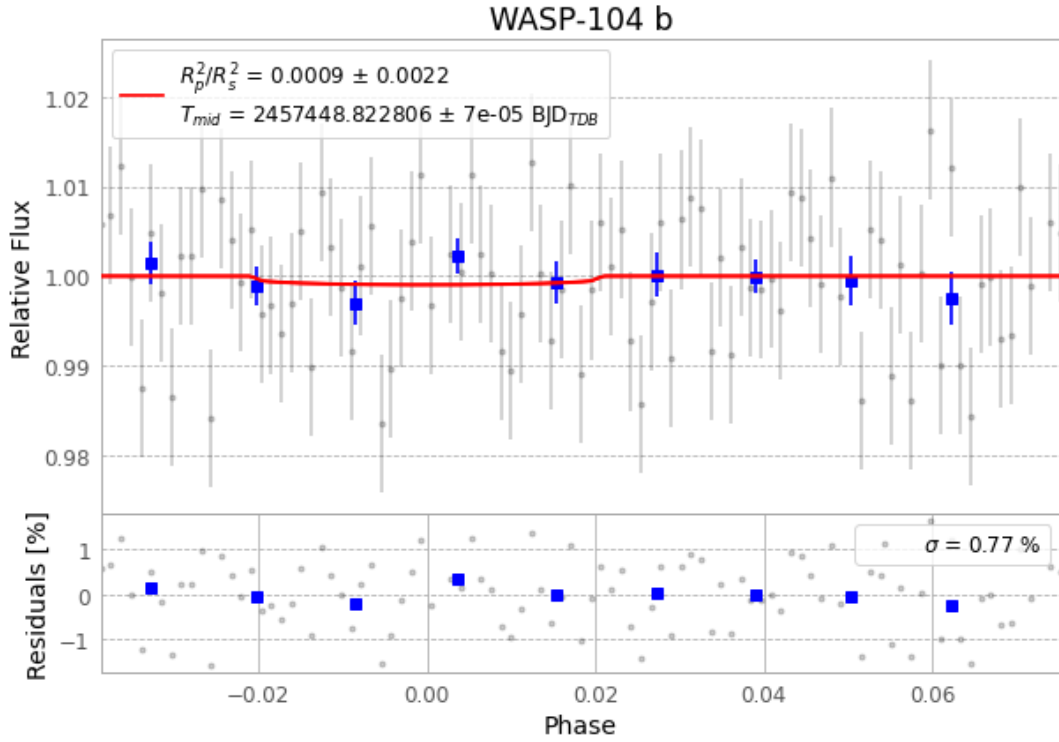


Figure 1 — FinalLightCurve_WASP-104 b_2016-02-29.png (SHA-256 dcc0f98023ba662a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [161, 224], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 bff14a0ea02f1d00...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

97 FITS frames (64 MB), WASP-104160301055712.FITS → WASP-104160301104512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-104-160301.log (89f75225fa34...), FinalLightCurve_WASP-104 b_2016-02-29.png (dcc0f98023ba...), FinalLightCurve_WASP-104 b_2016-02-29.csv (21fbe6d01b88...)

Compute resources

Wall time 102.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-28 00:08Z.